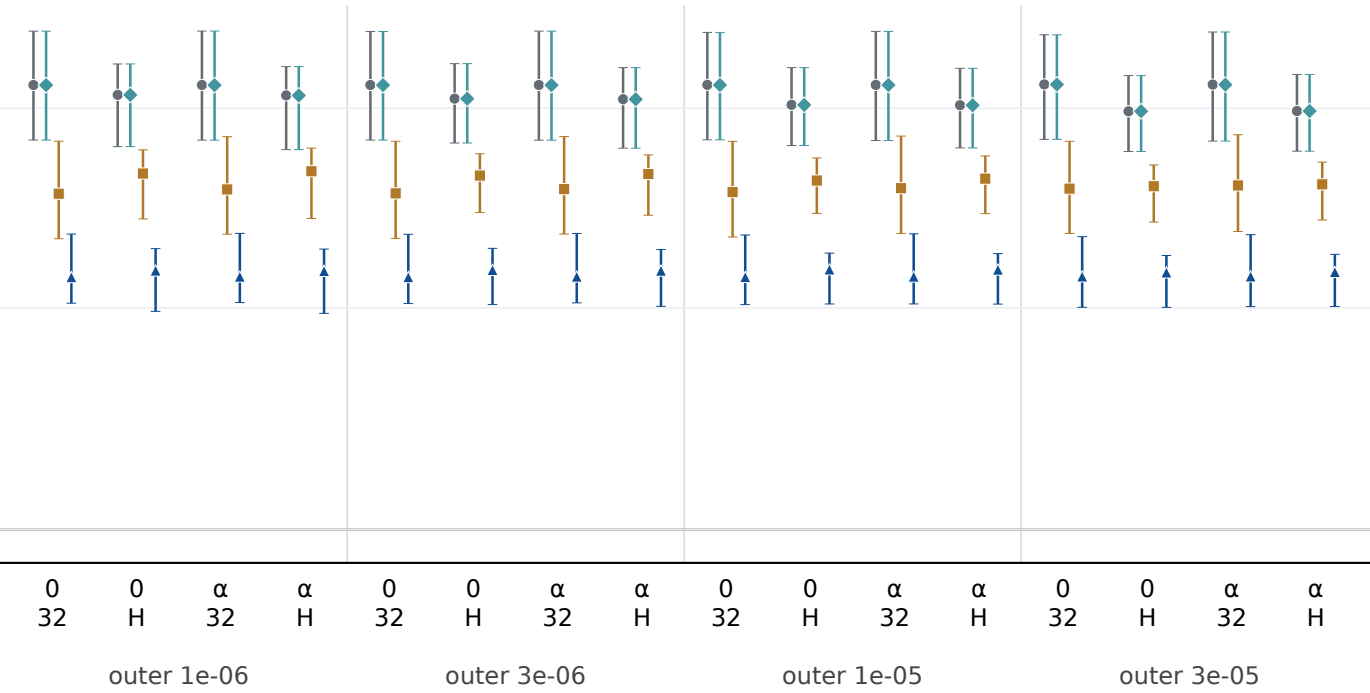


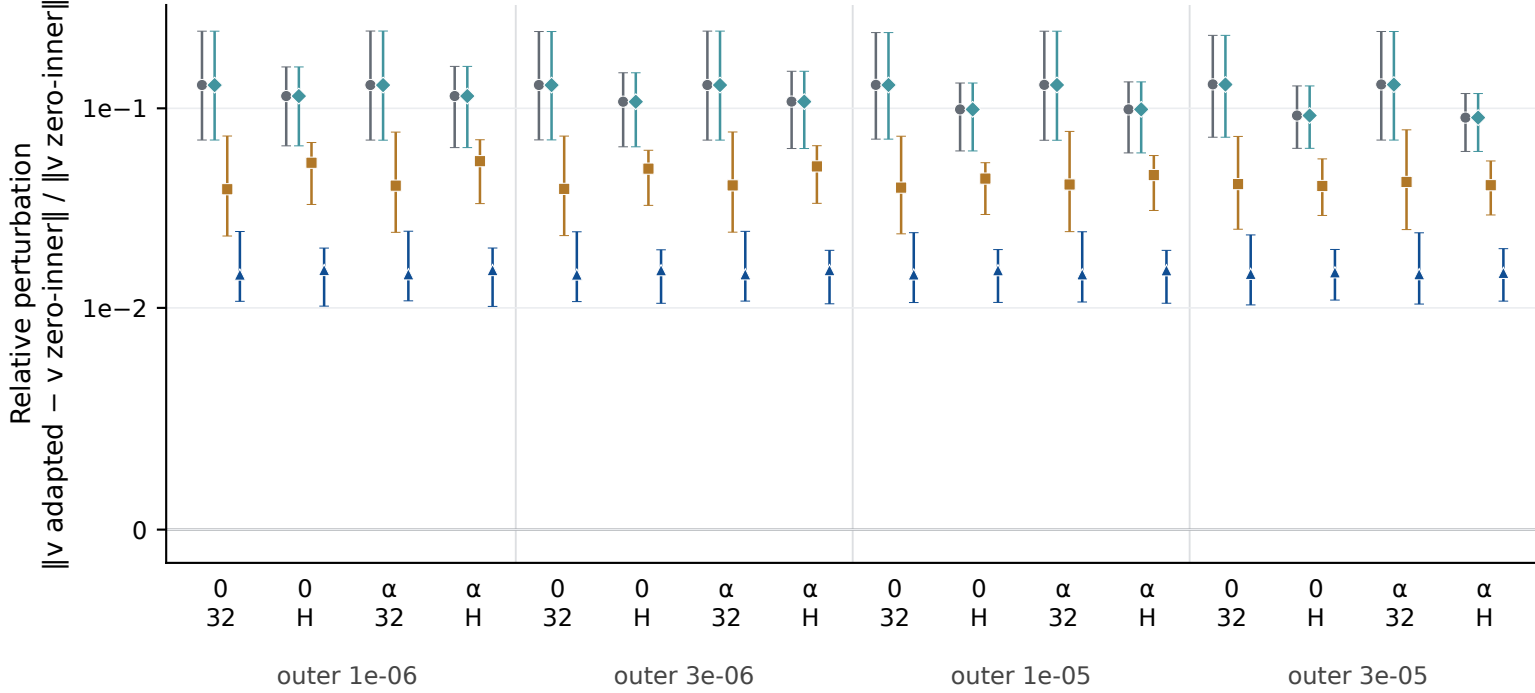
Paired query-field changes through AdamW

All 64 saved states · zero-inner versus inner $3e-5$ at identical weights and saved moments

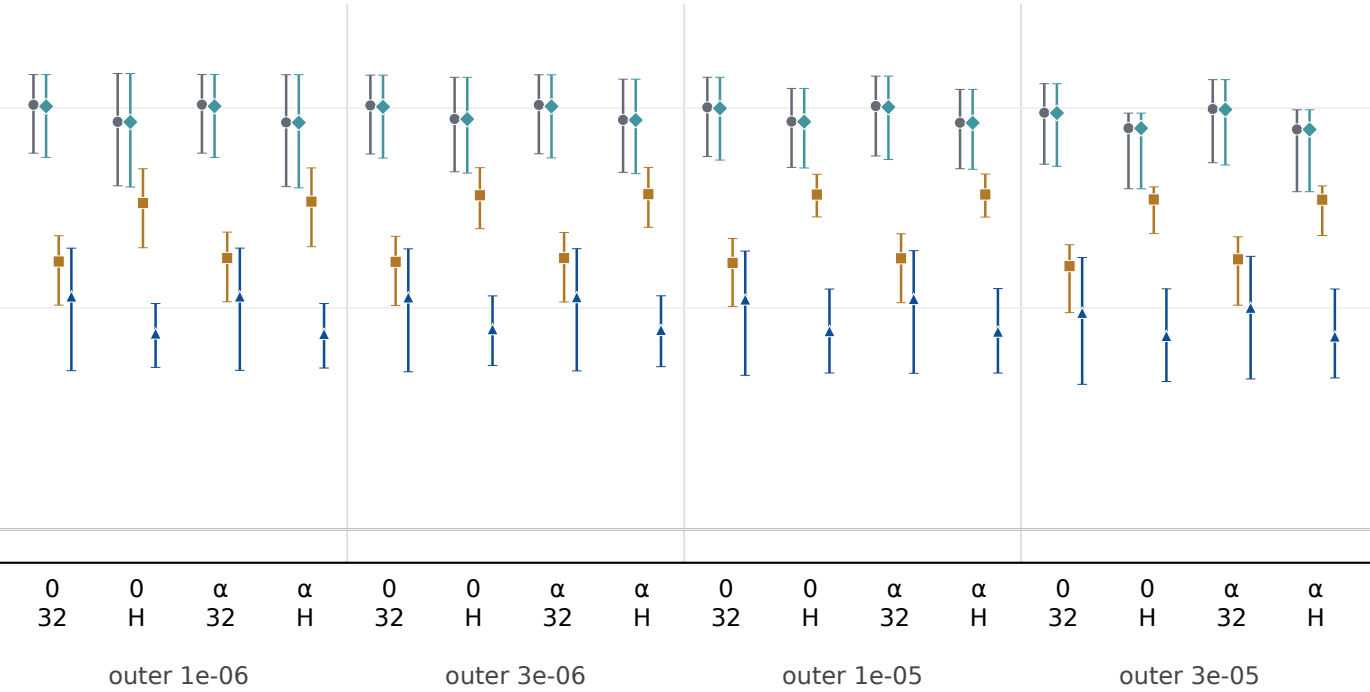
a Code · one pass



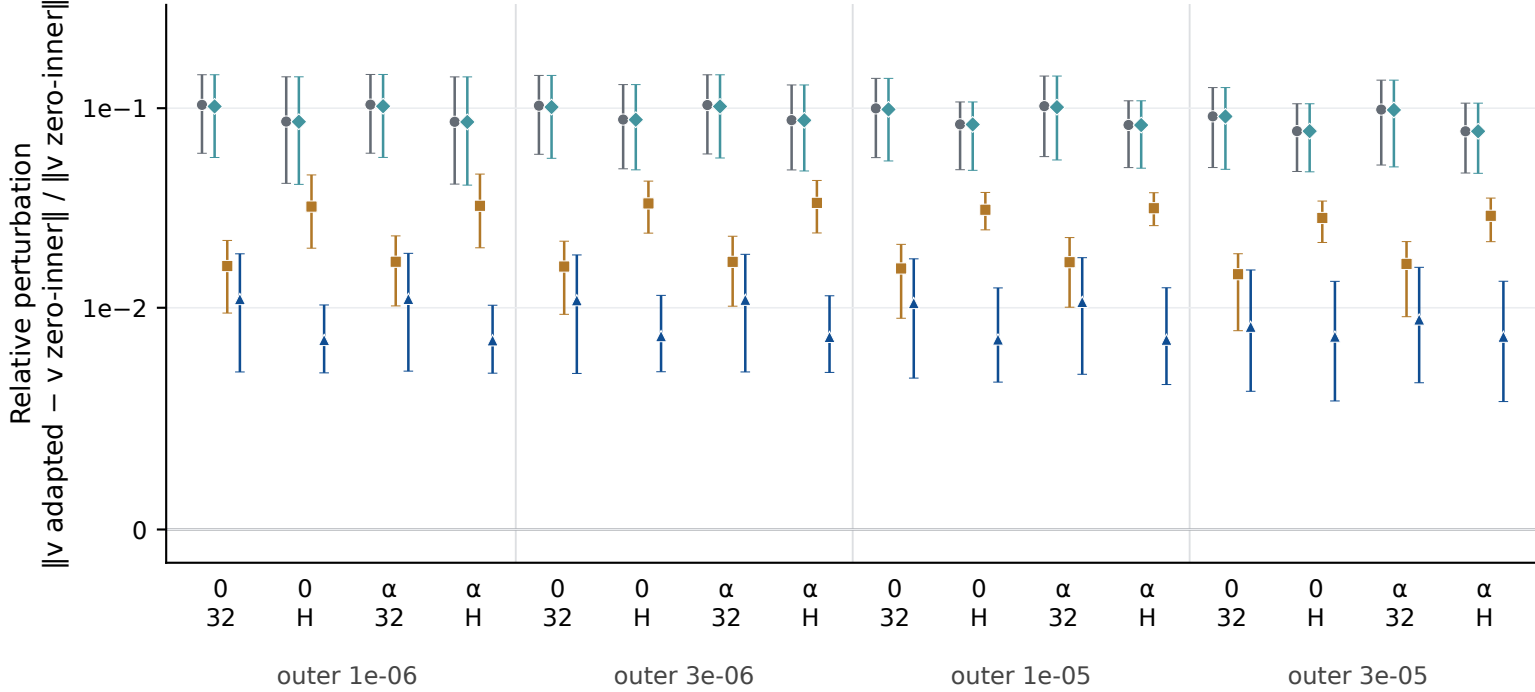
b Code · fixed quarter, four passes



c Math · one pass



d Math · fixed quarter, four passes



● Raw query gradient ◆ After global clipping ■ Updated first moment ▲ AdamW adaptive component

Markers: means; whiskers: minimum-maximum across all eight fixed TRAIN triples (not confidence intervals). Symmetric-log axes retain zeros; x with n/8 marks undefined or incomplete ratios. H denotes the terminal saved state. Stage denominators differ. Complete absolute perturbations and zero-reference norms are in all2048-stage-vectors.csv. One seed; dependent episodes, rates and checkpoints. Local vector changes do not establish trajectory mediation.